

CHAITANYA MITTAL

☎ (672) 336-1449 ✉ cma166@sfu.ca 🌐 /chaitanya-mittal-47ab8a2a4/ 📄 /ChaitanyaMittal27 📦 chaitanya-mittal.vercel.app/

Education

Simon Fraser University

Burnaby, BC

Bachelor of Computer Science — GPA: 3.81

Expected Graduation: Dec 2027

- Dean's Honour Roll — Summer 2024, Fall 2024, Spring 2025, Summer 2025

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, C/C++, HTML, CSS, SQL

Frontend: React.js, Next.js, Tailwind CSS, shadcn/ui, Three.js, Bootstrap

Backend: Node.js, Express.js, Flask, Spring Boot, REST APIs, SQLite, PostgreSQL, MongoDB, Supabase

ML & Data: scikit-learn, pandas, NumPy, Streamlit, Hugging Face Spaces

Other: Git, Postman, OAuth 2.0, JWT, Vercel, AWS EB, Cloudflare, Docker, Manifest v3, Figma

Technical Work Experience

Full Stack Developer

Blueprint SFU

Sep 2025 – Present

- Engineered a **3-step signup flow** using **React** and **Supabase Auth**, implementing email/password authentication, multi-field profile collection, and email verification to securely onboard member and admin users.
- Built a shared **React** component library (**shadcn/ui**, **Tailwind CSS**) with built-in validation and error handling, reused across **5+** form screens including signup, renewal, and profile editing flows.
- Implemented user deletion via **Supabase RPC** and **auth.admin.deleteUser** to fully remove records from both database and authentication layers, enforcing admin-only access control.

Projects

SFU Course Predictor | [Live Demo](#) | [Github](#)

scikit-learn | pandas | **GitHub Actions** | **Hugging Face**

- Trained a 3-model ML system (**Gradient Boosting + Random Forest**) on **48,900+** rows predicting course offering, capacity, and enrollment — achieving **AUC 0.864**, 74% accuracy, **R² 0.886–0.906**, and MAE under 15 seats/students on 1,522 unseen Spring 2026 courses.
- Engineered **20 temporal lookback features** across 19 semesters with strict no-leakage splitting, catching a critical data contamination bug where non-unique term indices caused all fall terms to leak into the test set.
- Built an automated semester refresh pipeline via **GitHub Actions** running Python scripts to collect, refit, and deploy updated models to **Hugging Face** with atomic rollback on failure.
- Integrated **Google Gemini** with a two-call architecture to parse free-text queries into structured model inputs, enabling a ChatGPT-style prediction interface alongside a manual input mode in **Streamlit**.

SFU Course Planner | [Live Demo](#) | [Github](#)

Spring Boot | Next | AWS | OAuth | JWT | PostgreSQL

- Built and deployed a production full-stack app using **Next.js** and **Spring Boot** enabling students to explore enrollment trends, visualize grade distributions, and compare courses across **500+** offerings and **50+** departments.
- Implemented **OAuth 2.0 PKCE** and **Supabase Auth** supporting **Google SSO** and email login, enabling users to create accounts, bookmark courses, and receive **daily enrollment alerts** via **EmailJS**.
- Reduced API response time from **7–8s to under 2s** by diagnosing slow query patterns and introducing structured indexing on course codes and department keys in **Supabase**.
- Designed a multi-cloud architecture (**Vercel**, **AWS EB**, **Supabase**, **Cloudflare**) with automated **24-hour** data refresh via **GitHub Actions**, integrating grade data and RateMyProfessors ratings alongside live SFU enrollment data.

EkantikSearch | [Live Demo](#) | [Github](#)

React | TypeScript | Flask | PostgreSQL | CI/CD

- Built a full-stack multilingual search engine using **React** and **Flask** that maps keyword queries to exact **YouTube timestamps**, returning direct video jump links across **800+** indexed videos.
- Automated a daily ingestion pipeline via **GitHub Actions** triggering Flask scripts to parse the **YouTube Data API**, extract timestamp-question mappings, with incremental updates, snapshot backups, and failure email alerts.
- Integrated **Azure Translator API** for Hindi-to-English transliteration, enabling romanized queries with **40% faster** retrieval after introducing **SQLite indexing**.